# Proprietary vs. Open-Source AI

## What Is Proprietary AI?

Proprietary (closed-source) AI is developed by a corporation whose model weights, training data, and architecture are trade secrets. Users access it via a paid API or product; they cannot inspect, modify, or reproduce the model.

- Examples: GPT-4 (OpenAI), Gemini (Google), Claude (Anthropic), Grok (xAI).
- Pros: typically best-in-class performance, enterprise SLAs, safety filtering, managed infrastructure.
- Cons: vendor lock-in, no transparency, data privacy concerns (prompts may be used for training), ongoing subscription cost.

## What Is Open-Source AI?

Open-source AI releases model weights (and sometimes training code/data) publicly. Anyone can download, run, fine-tune, and deploy the model — including on their own hardware.

- Examples: Llama 3 (Meta), Mistral, Falcon, Gemma (Google), Phi (Microsoft).
- Pros: transparency, customisability, privacy (data stays on-premise), no API costs at scale, research reproducibility.
- Cons: requires ML expertise to deploy, performance may lag frontier proprietary models, no liability coverage, potential misuse.

> *Trend: open-source models are closing the performance gap — Llama 3 70B competes with GPT-3.5 on many benchmarks.*

## Comparison Table

| Criterion | Proprietary | Open-Source |
|---|---|---|
| Access | API / product only | Download & run locally |
| Performance | State-of-the-art | Catching up fast |
| Transparency | Black box | Inspectable weights |
| Data Privacy | Prompts may be logged | Full local control |
| Cost | Pay per token | Free (compute costs) |
| Customisation | Limited (fine-tune API) | Full fine-tuning access |
| Safety filters | Built-in | User responsibility |

## Quick-Revision Questions

- What does 'open-source' mean in the context of AI models specifically?
- For a healthcare company with strict data privacy requirements, which would you recommend and why?
- Name two proprietary and two open-source LLMs.